

# School of Computer Science Engineering and Technology

## Lab: 1

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|-------------|-----------------------------|
| Course      | B. Tech.                    |
| Type        | Elective                    |
| Course Code | CSET346                     |
| Course Name | NATURAL LANGUAGE PROCESSING |
| Year        | 2026                        |
| Semester    | Odd                         |
| Batch       | 2026-27                     |

### CO-Mapping

| Question | C01 | C02 | C03 |
|----------|-----|-----|-----|
| Q1       | √   |     |     |
| Q2       |     | √   |     |
| Q3       |     | √   | √   |
| Q4       |     |     | √   |

### NLP Lab – Text Preprocessing and N-gram Analysis

#### Objective:

This lab aims to introduce students to Tokenization, Stop Words, Stemming, Lemmatization, Spelling Correction, and N-grams using Python and standard NLP libraries.

#### Problem Statement

Natural language text contains punctuation, common words, different grammatical forms, spelling errors, and sequential word patterns. Use suitable Python libraries and datasets to preprocess textual data and analyze word sequences.

#### Datasets / Resources

- NLTK Brown Corpus – standard English language corpus.
- NLTK Gutenberg Corpus – collection of literary texts.
- NLTK Words Corpus – vocabulary resource for spelling correction.

#### Instructions

- Use Python with NLTK, pandas, NumPy, or other suitable libraries.
- Download required NLTK resources before execution.
- Display intermediate and final outputs.
- Include clear comments in the code.

#### Q1. Use the NLTK Brown Corpus to perform sentence tokenization, word tokenization, and stop-word removal.

- (a) Load the Brown Corpus and select at least 20 sentences.
- (b) Perform sentence and word-level tokenization.
- (c) Convert tokens to lowercase.
- (d) Remove punctuation and English stop words.

(e) Display original and cleaned tokens for at least five sentences.

Expected Output: Original sentences, sentence tokens, word tokens, and cleaned tokens after stop-word removal.

**Q2. Apply stemming and lemmatization to words extracted from the Brown Corpus and compare the results.**

(a) Extract at least 30 alphabetic words.

(b) Apply Porter Stemmer.

(c) Apply WordNetLemmatizer.

(d) Include words such as playing, studies, running, better, and cars.

(e) Create a DataFrame containing original, stemmed, and lemmatized words.

Expected Output: A comparison table showing original words, stems, and lemmas.

**Q3. Implement basic spelling correction using the NLTK Words Corpus.**

(a) Load the NLTK Words Corpus.

(b) Create at least 10 intentionally misspelled words.

(c) Use approximate string matching to find corrections.

(d) Display misspelled words and suggestions.

(e) Correct at least five example sentences.

Expected Output: Misspelled words, suggested corrections, and corrected sentence examples.

**Q4. Use the NLTK Brown Corpus to generate and analyze unigram, bigram, and trigram sequences.**

(a) Load a suitable subset of the Brown Corpus.

(b) Retain lowercase alphabetic tokens.

(c) Generate unigrams, bigrams, and trigrams.

(d) Calculate N-gram frequencies.

(e) Display the ten most frequent unigrams, bigrams, and trigrams.

(f) Explain how increasing N captures more contextual information.

Expected Output: Preprocessed text, N-gram sequences, frequency tables, and the ten most frequent N-grams.